Mth634 midterm grand quiz

Lec (1-22) topic (1-85)

Due Date 2july2021

- Mth-634 Mega file solved by VU fellows 12july2020 pg198
- Mth634 BIG FILE WIKI 13july2020 (pg701)
- Mth634.A.S correct Solution 13july2020 (pg701)
- Mth634 Searching Grand Quiz by Sumera Skhan 3jan2021 pg111
- Mth634 Grand Quiz by AR (unsolve) 3jan2021 pg297
- Document from_GD Pilot WaRraiCh 634 (solve)3jan2021 pg297
- Mth634 SEARCHING Grand Quiz File By SUMERA 3jan2021 pg281
- Mth634 solve grand Quiz 2july2021 pg55
- Mth634 solve grand Quizz 2july2021 pg140

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Question # 15 of 30 (Start time: 09:37:54 PM, 03 January 2021)

Which of the following is the basis for a lower limit topology on R (The Set of Real Numbers)?

Select the correct option

○	$B_l = \{(a,b)\,\|a,b \in R\}$
○	$B_l = \{[a,b]\,\|a,b \in R\}$
■	$B_l = \{[a,b)\,\|a,b \in R\}$
○	$B_l = \{(a,b]\,\|a,b \in R\}$

S KHAN ACADEMY

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Question # 23 of 30 (Start time: 09:44:41 PM, 03 January 2021) Total Marks: 1

A map 'f' from a topological space X to a topological space Y is ..... if and only if the image of each element of a basis for TX(Topology on X) is open subset of Y.

Select the correct option

- injective
- an open map
- surjective
- bijective

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MTH634:Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 22 of 30 (Start time: 09:19:08 AM, 02 July 2021)

Total Marks: 1

The neighborhood system of a point p in an indiscrete space X is

Select the correct option

Reload Math Equations

- p
- $\emptyset$
- X
- none

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Question # 24 of 30 (Start time: 03:44:15 PM, 02 July 2021)

Total Marks: 1

Which of the following statement is true?

Select the correct option

Reload Math Equations

- all of the above
- $The\ collection\ B = \{(a, b] \subseteq \mathbb{R} : a < b\}\ of\ open\ intervals\ is\ a\ basis\ on\ \mathbb{R}.$
- $The\ collection\ B = \{(a, b) \subseteq \mathbb{R} : a < b\}\ of\ open\ intervals\ is\ a\ basis\ on\ \mathbb{R}.$ ✓
- $The\ collection\ B = \{[a, b) \subseteq \mathbb{R} : a < b\}\ of\ open\ intervals\ is\ a\ basis\ on\ \mathbb{R}.$

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MTH634:Grand Quiz

Quiz Start Time: 03:00 PM, 02 July 2021

Question # 15 of 30 (Start time: 03:12:30 PM, 02 July 2021)

Total Marks: 1

Let A be a non empty proper subset of an indiscrete topological space X. The exterior of A is

Select the correct option

Reload Math Equations

- A itself
- Ø

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sec(s)

MTH634:Grand Quiz

Quiz Start Time: 03:00 PM, 02 July 2021

Question # 19 of 30 (Start time: 03:16:19 PM, 02 July 2021)

Total Marks: 1

The basis consisting of all _____ in $\mathbb{R}^2$ generates the usual topology on $\mathbb{R}^2$.

Select the correct option

Reload Math Equations

- open balls
- closed balls

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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 20 of 30 (Start time: 05:21:05 PM, 02 July 2021)

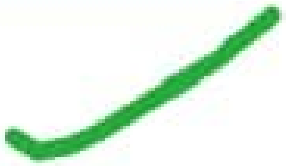
Total Marks: 1

Let A be a non empty proper subset of an indiscrete topological space X. The boundary of A is

Select the correct option

Reload Math Equations

○	$\emptyset$
○	X



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MTH634:Grand Quiz

Question # 16 of 30 (Start time: 04:56:18 PM, 02 July 2021)

Finite Complement topology defined on a set "A" consist of those subsets "U" of "A" such that either Uc is finite or ___

Select the correct option

○	U complement is infinite
○	U=Empty Set
○	U is infinite
●	U=A



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MTH634:Grand Quiz

Quiz Start Time: 08:17 PM, 02 July 2021

Question # 5 of 30 (Start time: 08:23:15 PM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d\}$. *The following set represents a topology on* X.

Select the correct option

Reload Math Equations

○	{φ,{a},{b},X}
○	{φ,{c},{a,b},X}
○	{φ,{a},{a,b},X}
○	{φ,{a},{b,c},X}

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Question # 24 of 30 (Start time: 08:44:22 PM, 02 July 2021)

Total Marks: 1

The set of all open intervals of $\mathbb{R}$ is a topology on $\mathbb{R}$, called

Select the correct option

Reload Math Equations

○	discrete topology
○	real topology
○	cofinite topology
○	usual topology ✓

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Question # 2 of 30 (Start time: 11:51:33 AM, 02 July 2021)

In a topological space X and a subset A of X , A is dense in X if and only if.....

Select the correct option

- $\bar{X} = X$
- $\bar{A} = A$
- $\bar{A} \subset X$
- $\bar{A} = X$ ✓

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MTH634:Grand Quiz

Quiz Start Time: 10:45 AM, 02 July 2021

Question # 4 of 30 (Start time: 10:47:54 AM, 02 July 2021)

Total Marks: 1

In a topological space X, let a set G is a neighborhood of each of its point. Then G is

Select the correct option

Reload Math Equations

- open
- closed

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MTH634:Grand Quiz

Quiz Start Time: 10:04 AM, 02 July 2021

Question # 9 of 30 (Start time: 10:11:13 AM, 02 July 2021)

Total Marks: 1

Let τ be a topology on X. The elements of τ are called:

Select the correct option

Reload Math Equations

○	closed set
○	dense set
○	open sets
●	derived set

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MTH634:Grand Quiz

Quiz Start Time: 11:50 AM, 02 July 2021

Question # 25 of 30 (Start time: 12:14:19 PM, 02 July 2021)

Total Marks: 1

Which of the following represents an open interval on real line R?

Select the correct option

Reload Math Equations

- $\{x|a < x < b\}$ *where* $a, b \in R$ *with usual order* $a < b$
- $\{x|a \leq x < b\}$ *where* $a, b \in R$ *with usual order* $a < b$
- $\{x|a \leq x \leq b\}$ *where* $a, b \in R$ *with usual order* $a < b$
- $\{x|a < x \leq b\}$ *where* $a, b \in R$ *with usual order* $a < b$

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MTH634:Grand Quiz

Question # 18 of 30 (Start time: 01:33:39 PM, 02 July 2021)

Union of topologies on a set X is :

Select the correct option

- may or may not be a topology
- none
- a topology
- not a topology

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Question # 25 of 30 (Start time: 02:03:10 PM, 02 July 2021)

In a topological space, the interior of a set A is the

Select the correct option

- largest open subset of A
- smallest open subset of A

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MTH634:Grand Quiz

Question # 14 of 30 (Start time: 03:58:02 PM, 02 July 2021)

Let (X, τ) be the topological space and $A, B \subset X$.If $A \subset B$ then:

Select the correct option

- $A \subset B$
- $\bar{A} \subset \bar{B}$ ✓
- None of above
- $\bar{A} \subset B$

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Question # 30 of 30 (Start time: 09:25:32 AM, 02 July 2021) Total Marks: 1

Topology means the study of something with respect to its .....

Select the correct option

- volume
- area
- perimeter
- place



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MTH634:Grand Quiz

Quiz Start Time: 05:39 PM, 02 July 2021

Question # 6 of 30 (Start time: 05:43:34 PM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d\}$. *The following set is a topology on* X.

Select the correct option

Reload Math Equations

- {φ,{a},{b},{c},X}
- {φ,{c,d},{b,c,d},X}
- none
- {φ,{a},{b},X}

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MTH634:Grand Quiz

Quiz Start Time: 08:17 PM, 02 July 2021

Question # 13 of 30 (Start time: 08:31:59 PM, 02 July 2021)

Total Marks: 1

Let $\mathbb{R}$ be real number. The following statement is true.

Select the correct option

Reload Math Equations

- The class δ of all infinite close intervals is a subbase for $\mathbb{R}$.
- The class δ of all infinite open intervals is a subbase for $\mathbb{R}$.



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MTH634:Grand Quiz

Quiz Start Time: 08:20 PM, 02 July 2021

Question # 23 of 30 (Start time: 08:44:24 PM, 02 July 2021) Total Marks: 1

$$\overline{(\bar{A})} =$$

Select the correct option

Reload Math Equations

- A
- None of above
- $\bar{A}$
- ϕ

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MTH634 Grand Quiz

Quiz Start Time: 08:25 PM, 02 July 2021

Question # 1 of 30 (Start time: 08:25:14 PM, 02 July 2021)

Let $X = \{2, 4, 5, 7\}$. *The following set represents a topology on* X.

Select the correct option

Reload Math Equations

- $\{\varphi, \{2\}, \{4, 5, 7\}, X\}$
- $\{\varphi, \{2\}, \{4\}, \{5\}, X\}$
- $\{\varphi, \{2\}, \{4, 5\}, X\}$
- $\{\varphi, \{2\}, \{5, 7\}, X\}$

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MTH634:Grand Quiz

Quiz Start Time: 08:20 PM, 02 July 2021

Question # 17 of 30 (Start time: 08:44:03 PM, 02 July 2021)

Total Marks: 1

The boundary of the four intervals $[a, b]$, (a, b), $(a, b]$, $[a, b)$ is

Select the correct option

Reload Math Equations

- {a, b}
- (a, b)



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sec(s)

MTH634:Grand Quiz

Quiz Start Time: 06:56 PM, 02 July 2021

Question # 30 of 30 (Start time: 08:26:21 PM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d\}$. *The following set is a topology on* X.

Select the correct option

Reload Math Equations

○	{φ,{b,d},{a,b,c},{b},X}
○	{φ,{b},{a,b},{c},X}
○	none
○	{φ,{b},{c},X}

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MTH634:Grand Quiz

Quiz Start Time: 08:20 PM, 02 July 2021

Question # 9 of 30 (Start time: 08:36:25 PM, 02 July 2021)

Total Marks: 1

Let $X = \{1, 2, 3\}$ and $\tau = \{\phi, \{1\}, \{2\}, \{1, 2\}, X\}$ then (X, τ) as a pair will be called .....

Select the correct option

Reload Math Equations

- Discrete topology
- Indiscrete topology
- Topological space
- None of these

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MTH634:Grand Quiz

Quiz Start Time: 08:20 PM, 02 July 2021

Question # 7 of 30 (Start time: 08:33:47 PM, 02 July 2021)

Total Marks: 1

Let $\{T_i | i \in I\}$ be any collection of topologies on set X. Then $\bigcap_{i \in I} T_i$ is :

Select the correct option

Reload Math Equations

- not a topology
- a topology
- empty
- empty set

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MTH634:Grand Quiz

Quiz Start Time: 08:20 PM, 02 July 2021

Question # 12 of 30 (Start time: 08:30:55 PM, 02 July 2021) **Total Marks:** 1

Let $X = R$ with usual topology and $A = (0, 4)$ then the limit point of A is:

Select the correct option

Reload Math Equations

- -2
- 2
- 5
- 6

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MTH634:Grand Quiz

Quiz Start Time: 10:45 AM, 02 July 2021

Question # 6 of 30 (Start time: 10:50:37 AM, 02 July 2021)

Total Marks: 1

Let $X = \mathbb{R}^2$ with usual topology and $A = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < 4\}$. *Closure of A is*

Select the correct option

Reload Math Equations

- $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 4\}$
- $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \geq 4\}$
- $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 4\}$ ✓
- $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 > 4\}$

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Question # 8 of 30 (Start time: 08:27:20 PM, 02 July 2021) Total Marks: 1

More than one topological spaces can be defined on a single set.

Select the correct option

- False
- True



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Question # 4 of 30 (Start time: 10:06:38 AM, 02 July 2021)

Total Marks: 1

K-Topology on the real line is not finer than the usual topology.

Select the correct option

- True
- False

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Question # 8 of 30 (Start time: 10:10:28 AM, 02 July 2021)
Total Marks: 1

Interior of a set A(denoted by Ao) which is a subset of a topological space X is the .....of all ..... subsets of X contained in A.

Select the correct option

- intersection, closed
- union, open
- union,closed
- intersection, open

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Question # 11 of 30 (Start time: 09:10:26 AM, 02 July 2021) Total Marks: 1

The trivial topology is also called:

Select the correct option

- Indiscrete topology
- Discrete topology
- Complete topology
- none

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Question # 27 of 30 (Start time: 08:24:32 PM, 02 July 2021)
Total Marks: 1

If one shape can be deformed in another shape then topologically they are considered to be .....

Select the correct option

- dissimilar
- partially equal
- unique
- the same

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Question # 17 of 30 (Start time: 08:35:22 PM, 02 July 2021)

Total Marks: 1

Let (X, τ) be the topological space and $A, B \subset X$ such that $A \subset B$ then for derived set

$$A'$$

and

$$B'$$

Select the correct option

Reload Math Equations

- $(A \cup B)' = A' \cup B'$
- $(A \cup B)' = A' \cap B'$
- None of above
- $(A \cup B)' = A \cup B$

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Question # 3 of 30 (Start time: 08:22:08 PM, 02 July 2021)

Total Marks: 1

A map ‘f’ from a topological space X to a topological space Y is an open map if and only if the image of …….

Select the correct option

- one open subset of range space is open in domain space
- each open subset of range space is open in domain space
- each open subset of domain space is open in range space ✓
- no open subset of domain space is open in range space

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Question # 6 of 30 (Start time: 08:31:25 PM, 02 July 2021) Total Marks: 1

For a function f: $X \to Y$, the restriction of f to A where A is a subset of X is continuous if f: $X \to Y$ is .....

Select the correct option

- discontinuous
- none of these
- piecewise continous
- continuous ✓

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MTH634:Grand Quiz

Question # 17 of 30 (Start time: 10:18:04 AM, 02 July 2021)

Consider the following topology on $X = \{a, b, c, d, e\}$

$$T = \{\emptyset, X, \{a\}, \{a, b\}, \{a, c, d\}, \{a, b, c, d\}, \{a, b, e\}\}$$

The member of the relative topology on $A = \{a, b\}$ is

Select the correct option

- ○ {c}
- ○ {e}
- ○ {a, b} ✓
- ○ none

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Question # 10 of 30 (Start time: 10:24:32 AM, 02 July 2021)
Total Marks: 1

If B is a basis for a topology T on a set X then ..... element of T can be written as union of the elements of B.

Select the correct option

- no
- every
- one
- some

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MTH634:Grand Quiz

Quiz Start Time: 10:16 AM, 02 July 2021

Question # 12 of 30 (Start time: 10:25:47 AM, 02 July 2021)

Total Marks: 1

$$Let\ X\ be\ a\ set,\ and\ let\ B = \{\{x\} : x \in X\}.\ Which\ of\ the\ following\ statement\ is\ true?$$

Select the correct option

Reload Math Equations

- *B* is not a basis for the topology on *X*
- *B* is a basis for the topology on *X*



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MTH634:Grand Quiz

Quiz Start Time: 10:04 AM, 02 July 2021

Question # 1 of 30 (Start time: 10:04:12 AM, 02 July 2021)

Total Marks: 1

For an open ball centered at x=(x1,x2,...xn) belongs to Rn, which of the following is the correct representation?

Select the correct option

Reload Math Equations

- $\{y|\sqrt{\sum_{i=1}^{n}(x_i-y_i)^2} \leq r\} \subset R^n, where\, y=(y_1,y_2,\ldots,y_n) \in R^n$
- $\{y|\sqrt{\sum_{i=1}^{n}(x_i-y_i)^2} \geq r\} \subset R^n, where\, y=(y_1,y_2,\ldots,y_n) \in R^n$
- $\{y|\sqrt{\sum_{i=1}^{n}(x_i-y_i)^2} > r\} \subset R^n, where\, y=(y_1,y_2,\ldots,y_n) \in R^n$
- $\{y|\sqrt{\sum_{i=1}^{n}(x_i-y_i)^2} < r\} \subset R^n, where\, y=(y_1,y_2,\ldots,y_n) \in R^n$

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Question # 27 of 30 (Start time: 10:08:32 AM, 02 July 2021) Total Marks: 1

If B is a basis for a topology T on a set X then every element of T can be written as ..... of the elements of B.

Select the correct option

- difference
- complement
- union
- intersection

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Question # 22 of 30 (Start time: 11:09:18 AM, 02 July 2021) **Total Marks: 1**

Consider the following topology on $X = \{a, b, c, d, e\}$:

$$T = \{\emptyset, X, \{a\}, \{a, b\}, \{a, c, d\}, \{a, b, c, d\}, \{a, b, e\}\}$$

The neighborhood of the point e is

Select the correct option

Reload Math Equations

- {a, b, c}
- {a, b, c, e}
- none
- {a, b}

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MTH634:Grand Quiz

Quiz Start Time: 11:50 AM, 02 July 2021

Question # 1 of 30 (Start time: 11:50:41 AM, 02 July 2021)

Total Marks: 1

In a topological space the intersection of any collection of closed sets is .....

Select the correct option

- Open
- None of these
- Neither open nor closed
- Closed

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Question # 5 of 30 (Start time: 11:56:49 AM, 02 July 2021) Total Marks: 1

The largest topology defined on some set is the .....topology.

Select the correct option

- ○ lower limit
- ● discrete
- ○ co finite
- ○ indiscrete

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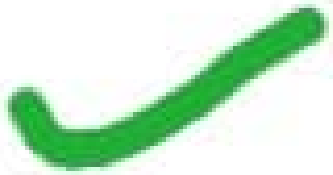
Question # 30 of 30 (Start time: 08:48:46 PM, 02 July 2021)

Total Marks: 1

Topology can be a useful tool in those problems where .....study is more effective.

Select the correct option

- Quantitative
- Qualitative



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Question # 8 of 30 (Start time: 12:00:22 PM, 02 July 2021) Total Marks: 1

A function whose range is a singleton is called a ..... function.

Select the correct option

- continuous
- Bijective
- one to many
- constant



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Question # 14 of 30 (Start time: 12:04:45 PM, 02 July 2021) Total Marks: 1

For a function f: $X \rightarrow Y$, the restriction of f to A where A is a subset of X is a function $fA:A \rightarrow Y$ defined as .....

Select the correct option

- $fA(a) \subseteq f(a)$ for all a belongs to A.
- $fA(a)=f(y)$ for all y belongs to Y.
- $fA(a)=f(a)$ for all a belongs to A. ✓
- $fA(a) \subseteq f(y)$ for all y belongs to Y.

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/TH634:Grand Quiz

uestion # 1 of 30 (Start time: 12:04:44 PM, 02 July 2021)

Which of the following is the basis for an upper limit topology on R (The Set of Real Numbers)?

elect the correct option

- $B_{up} = \{(a, b] \,|a, b \in R\}$ ✓
- $B_{up} = \{[a, b] \,|a, b \in R\}$
- $B_{up} = \{(a, b) \,|a, b \in R\}$
- $B_{up} = \{[a, b) \,|a, b \in R\}$

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MTH634:Grand Quiz

Quiz Start Time: 11:50 AM, 02 July 2021

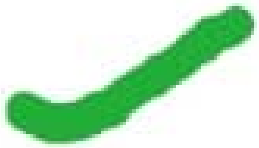
Question # 19 of 30 (Start time: 12:09:31 PM, 02 July 2021)

Total Marks: 1

If ζ is a subbase for topologies T and T^*, then

Select the correct option

Reload Math Equations

- $T = T^*$ 
- $T \neq T^*$

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Question # 9 of 30 (Start time: 12:12:28 PM, 02 July 2021) Total Marks: 1

In a discrete space X, the derived set A' of any subset A of X is ..... .

Select the correct option

- A complement
- empty set
- A itself
- X

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Question # 13 of 30 (Start time: 12:17:16 PM, 02 July 2021)
Total Marks: 1

If X has 4 elements then power set of X has ..... elements

Select the correct option

- ○ 8
- ○ 4
- ○ 2
- ○ 16 

Click to Save Answer & Move to Next Question

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Question # 27 of 30 (Start time: 12:16:43 PM, 02 July 2021)

Total Marks: 1

The set of all singletons forms a basis for ..... topology.

Select the correct option

- co finite
- indiscrete
- lower limit
- discrete



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Question # 16 of 30 (Start time: 12:06:45 PM, 02 July 2021)
Total Marks: 1

The set of .....of R (Real line) forms a topology called usual topology.

Select the correct option

- all open intervals
- all open balls
- all open spheres
- all open discs

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Question # 12 of 30 (Start time: 12:16:31 PM, 02 July 2021) Total Marks: 1

Closure of a subset A of a topological space X is the intersection of all ..... of X containing A.

Select the correct option

- subsets
- closed subsets
- open subsets
- none of these

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Question # 18 of 30 (Start time: 12:23:14 PM, 02 July 2021)

A subset A of a topological space X is closed if and only if .....

Select the correct option

○	$A' = A^c$
○	$A = A^c$
○	$A \supset A'$
○	$A' \subset A$

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Question # 28 of 30 (Start time: 12:18:13 PM, 02 July 2021)

Total Marks: 1

Only one topology can be defined on a set.

Select the correct option

- False 
- True

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Question # 16 of 30 (Start time: 12:20:49 PM, 02 July 2021) Total Marks: 1

Which of the following topology is called “Finite Complement Topology”?

Select the correct option

- Discrete Topology
- Indiscrete Topology
- Co finite Topology
- Lower Limit Topology

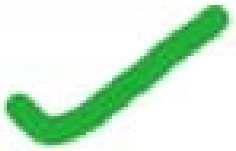
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Question # 3 of 30 (Start time: 05:41:19 PM, 02 July 2021)
Total Marks: 1

Derived set of empty set .....

Select the correct option

- All of above
- 1
- Non empty
- Empty



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MTH634:Grand Quiz

Question # 17 of 30 (Start time: 12:21:53 PM, 02 July 2021)

Let τ_1 and τ_2 be the two topologies on a non-empty set X. then τ_2 is finer than τ_1 iff.....

Select the correct option

○	$\tau_1 = \tau_2$
○	$\tau_2 \subseteq \tau_1$
○	$\tau_1 \subseteq \tau_2$ 
○	None of these

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MTH634:Grand Quiz

Question # 19 of 30 (Start time: 12:24:04 PM, 02 July 2021)

Which of the following is true for a subset A of a Topological space X.

Select the correct option

○	$X = A^0 \cup Ext(A) \cup bd(A)$ ✓
○	$X = A^0 \cap Ext(A) \cup bd(A)$
○	$X = A^0 \cup Ext(A) \cap bd(A)$
○	$X = A^0 \cap Ext(A) \cap bd(A)$

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Question # 20 of 30 (Start time: 12:24:51 PM, 02 July 2021)
Total Marks: 1

Which of the following are NOT topologically equal?

Select the correct option

- A triangle and a rectangle
- A rectangle and a square
- A circle and a line 
- A circle and a square

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Question # 22 of 30 (Start time: 12:26:46 PM, 02 July 2021)

Total Marks: 1

Which of the following statement is true?

Select the correct option

- In both Geometry and Topology we care about exact measurements.
- In Geometry we don't care about measurements but in Topology we care about exact measurements.
- In Geometry we care about measurements but in Topology we don't care about exact measurements.
- In both Geometry and Topology we don't care about exact measurements.

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estion # 23 of 30 (Start time: 12:27:50 PM, 02 July 2021)

Let $X = \{a, b, c, d\}$ *and* $\tau = \{\phi, \{c\}, \{a, c\}, \{b, c, d\}, X\}$ *be a topology on* X. The closed set in X is:

lect the correct option

○	{c}
○	{d}
○	none
○	{b,d}

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Question # 6 of 30 (Start time: 01:05:56 PM, 02 July 2021)
Total Marks: 1

A subset U of R2 is open in R2 iff it contains an ..... at each of its points.

Select the correct option

- (●) open disk
- ( ) closed disk
- ( ) closed interval
- ( ) open sphere

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Question # 13 of 30 (Start time: 01:10:18 PM, 02 July 2021)

Total Marks: 1

Every collection of subsets of a set is said to be a topology.

Select the correct option

- ● False
- ○ True

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Question # 18 of 30 (Start time: 01:14:01 PM, 02 July 2021)
Total Marks: 1

Exterior point of a set is the ..... point of its complement.

Select the correct option

- limit
- interior 
- exterior
- boundary

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Question # 27 of 30 (Start time: 01:22:48 PM, 02 July 2021)

Total Marks: 1

Let X={1,2,3}, then P(X)=_____

Select the correct option

Reload Math Equations

- {φ,{1},{3},{1,2},{1,3},{1,2,3}}
- {φ,{1},{2},{3},{1,2},{1,3},{2,3},X} ✓
- {φ,X}
- {φ,{1},{2},{3},{1,2},{1,3},{2,3}}

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MTH634:Grand Quiz

Quiz Start Time: 12:59 PM, 02 July 2021

Question # 24 of 30 (Start time: 01:19:38 PM, 02 July 2021)

Total Marks: 1

Which of the following statement is true?

Select the correct option

Reload Math Equations

- For A, B Subset of a topological Space X, $If\ A \subset B\ then\ A' \supseteq B'$
- For A, B Subset of a topological Space X, $If\ A \subset B\ then\ A' \subset B'$ ✓
- For A, B Subset of a topological Space X, $If\ A \supset B\ then\ A' \subset B'$
- For A, B Subset of a topological Space X, $If\ A \subset B\ then\ A' = B'$

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MTH634:Grand Quiz

Quiz Start Time: 02:50 PM, 02 July 2021

Question # 4 of 30 (Start time: 02:52:32 PM, 02 July 2021)

Total Marks: 1

If in a topology τ on X, all subsets of X are called open and closed, then τ is called :

Select the correct option

Reload Math Equations

- discrete space
- none
- metric space
- indiscrete space

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Question # 3 of 30 (Start time: 03:02:39 PM, 02 July 2021) Total Marks: 1

The lower limit topology is ..... than the usual topology.

Select the correct option

- finer
- none of these
- coarser
- equal

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MTH634:Grand Quiz

Quiz Start Time: 03:00 PM, 02 July 2021

Question # 2 of 30 (Start time: 03:02:03 PM, 02 July 2021)

Total Marks: 1

If X is finite and has n elements then power set of X has ____ elements.

Select the correct option

Reload Math Equations

○	none
●	2^n
○	2^{n+1}
○	2^{n-1}

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MTH634:Grand Quiz

Quiz Start Time: 02:50 PM, 02 July 2021

Question # 11 of 30 (Start time: 03:00:23 PM, 02 July 2021)

Total Marks: 1

For any set there are always two topologies namely.....
.....

Select the correct option

- Co finite and Indiscrete Topology
- Lower limit topology and discrete topology
- Discrete Topology and cofinite Topology
- Discrete and Indiscrete Topology



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MTH634:Grand Quiz

Quiz Start Time: 03:00 PM, 02 July 2021

Question # 5 of 30 (Start time: 03:04:04 PM, 02 July 2021)

Total Marks: 1

Let $X = \mathbb{R}$ with usual topology and consider $B = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\}$. *The limit point of* B is:

Select the correct option

Reload Math Equations

○	none
○	½
○	0
○	1

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MTH634:Grand Quiz

Quiz Start Time: 03:00 PM, 02 July 2021

Question # 7 of 30 (Start time: 03:05:19 PM, 02 July 2021)

Total Marks: 1

Let $X = \mathbb{R}$ with usual topology and $A = (0, 3)$. *The limit point of* A is:

Select the correct option

Reload Math Equations

○	-0.5
○	5
○	1.5
○	4

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MTH634:Grand Quiz

Quiz Start Time: 03:00 PM, 02 July 2021

Question # 8 of 30 (Start time: 03:06:01 PM, 02 July 2021)

Total Marks: 1

Let $X = \{2, 4, 6\}$. The following sets represents a topology on

$$X$$

Select the correct option

Reload Math Equations

○	$\{\phi, \{4\}, \{4, 6\}, X\}$
○	$\{\phi, X\}$
○	All of these.
○	$\{\phi, \{2\}, \{2, 4\}, X\}$

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sec(s)

MTH634:Grand Quiz

Quiz Start Time: 02:50 PM, 02 July 2021

Question # 21 of 30 (Start time: 03:07:37 PM, 02 July 2021)

Total Marks: 1

We have ..... topological spaces for a three point set.

Select the correct option

- 29
- 28
- 26
- 27

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MTH634:Grand Quiz

Quiz Start Time: 02:50 PM, 02 July 2021

Question # 23 of 30 (Start time: 03:09:21 PM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d, e\}$ with topology $T = \{\phi, \{b\}, \{a, d\}, \{a, b, d\}, \{a, c, d, e\}, X\}$. The interior point of $A = \{a, b, c\}$ is:

Select the correct option

Reload Math Equations

- d
- b
- c
- a

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Question # 12 of 30 (Start time: 03:10:13 PM, 02 July 2021) Total Marks: 1

Which of the following statement is not true ?

Select the correct option

○	Discrete topology is finer than the indiscrete topology defined on the same non empty set.
○	Indiscrete topology is weaker than any other topology defined on the same non empty set.
○	Indiscrete topology is finer than any other topology defined on the same non empty set.
○	Discrete topology is finer than any other topology defined on the same non empty set.

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Question # 13 of 30 (Start time: 03:10:55 PM, 02 July 2021)

Total Marks: 1

Which of the following is the basis for a K-Topology defined on real line where K is a subset of R?

Select the correct option

Reload Math Equations

- $B_k = \{(a,b] \,|a,b \in R\} \cup \{(a,b] \backslash K \,|a,b \in R\}$
- $B_k = \{[a,b) \,|a,b \in R\} \cup \{[a,b) \backslash K \,|a,b \in R\}$
- $B_k = \{(a,b) \,|a,b \in R\} \cup \{(a,b) \backslash K \,|a,b \in R\}$
- $B_k = \{[a,b] \,|a,b \in R\} \cup \{[a,b] \backslash K \,|a,b \in R\}$

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Question # 14 of 30 (Start time: 03:11:46 PM, 02 July 2021) Total Marks: 1

If X=R with usual Topology and consider B= {1/n where n belongs to the set of Natural numbers}, then the limit point of B is .....

Select the correct option

○	1
○	1/3
○	0
○	1/2



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MTH634:Grand Quiz

Quiz Start Time: 03:00 PM, 02 July 2021

Question # 16 of 30 (Start time: 03:13:09 PM, 02 July 2021)

Total Marks: 1

Subset A of a topological space (X, τ) is.....iff $\bar{A} = A$.

Select the correct option

Reload Math Equations

- Closed 
- open

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MTH634:Grand Quiz

Quiz Start Time: 03:00 PM, 02 July 2021

Question # 17 of 30 (Start time: 03:13:50 PM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d, e\}$. *The following set* is a *topology on* X.

Select the correct option

Reload Math Equations

○	none
○	{φ,{a},{a,b},{a,c,d},{a,b,c,d},{a,b,e},X}
○	{φ,{a},{b,c},{a,b,c},{d},X}
○	{φ,{a,b},{a,c,d},X}

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MTH634:Grand Quiz

Quiz Start Time: 03:00 PM, 02 July 2021

Question # 18 of 30 (Start time: 03:15:38 PM, 02 July 2021)

Total Marks: 1

Let $X = R$ be a usual topology and consider $C = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4} \ldots\ldots\ldots\ldots\ldots\}$ then which of the following is not limit point of :

Select the correct option

Reload Math Equations

○	0
○	1/2
○	Both (b) and (c)
○	1

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Question # 21 of 30 (Start time: 03:17:08 PM, 02 July 2021) Total Marks: 1

..... is the largest topology containing both T1 and T2.

Select the correct option

- T1 union T2
- Indiscrete topology
- Cofinite topology
- T1 intersection T2 

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MTH634:Grand Quiz

Quiz Start Time: 02:50 PM, 02 July 2021

Question # 30 of 30 (Start time: 03:16:50 PM, 02 July 2021)

Total Marks: 1

Let X be a topological space and A, a subset of X, then

Select the correct option

Reload Math Equations

- none
- $\bar{A} = A \cap A'$
- $\bar{A} = A \cup A'$
- $A' = A \cup \bar{A}$

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MTH634:Grand Quiz

Quiz Start Time: 03:00 PM, 02 July 2021

Question # 22 of 30 (Start time: 03:17:59 PM, 02 July 2021)

Total Marks: 1

$The\ collection\ \tau\ of\ subsets\ of\ X\ consisting\ of\ the\ empty\ set\ \phi\ and\ all\ subsets\ of\ X\ whose\ complements\ are\ finite\ is\ called:$

Select the correct option

Reload Math Equations

○	cofinite topology
○	discrete topology
○	indiscrete topology
○	none

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MTH634:Grand Quiz

Quiz Start Time: 03:00 PM, 02 July 2021

Question # 23 of 30 (Start time: 03:18:59 PM, 02 July 2021)

Total Marks: 1

$$\bigcap_{n \in N} \left(-\frac{1}{n}, \frac{1}{n} \right) = \ldots\ldots\ldots, \text{ where N stands for set of natural numbers}$$

Select the correct option

Reload Math Equations

○	{0}
○	(-infinity, +infinity)
○	{}
○	{-1,1}

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Question # 25 of 30 (Start time: 03:20:25 PM, 02 July 2021) Total Marks: 1

Which of the following is the topology on real line induced by the set of all linear functions from R to R with lower limit topology is .....

Select the correct option

- cofinite
- upper limit
- Indiscrete
- discrete



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MTH634:Grand Quiz

Quiz Start Time: 03:00 PM, 02 July 2021

Question # 26 of 30 (Start time: 03:21:44 PM, 02 July 2021)

Total Marks: 1

If τ is a topology on a non empty set X, then ..... of finite members of τ belongs to τ.

Select the correct option

Reload Math Equations

○	union
○	intersection



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MTH634:Grand Quiz

Quiz Start Time: 03:00 PM, 02 July 2021

Question # 27 of 30 (Start time: 03:22:51 PM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d\}$. *The following set* is not *a topology on* X.

Select the correct option

Reload Math Equations

○	{φ,{a},{a,b},X}
○	{φ,{a},{a,c},X}
○	{φ,{a},{b},X}
○	{φ,{a},{a,d},X}

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MTH634:Grand Quiz

Quiz Start Time: 02:30 PM, 02 July 2021

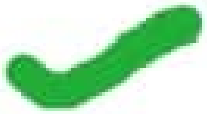
Question # 12 of 30 (Start time: 02:39:13 PM, 02 July 2021)

Total Marks: 1

The neighborhood system of a point p in an indiscrete space X is

Select the correct option

Reload Math Equations

- X 
- p
- $\emptyset$
- none

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Question # 2 of 30 (Start time: 03:22:22 PM, 02 July 2021) Total Marks: 1

Boundary point of a set is also called..... point.

Select the correct option

- exterior
- interior
- limit
- frontier



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Question # 28 of 30 (Start time: 03:23:29 PM, 02 July 2021) Total Marks: 1

The set of all open disks forms a usual basis for the usual topology defined on ……

Select the correct option

- Rn(n tuple)
- R(Real line)
- R2(Real plane) 
- R3(space)

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Question # 29 of 30 (Start time: 03:24:15 PM, 02 July 2021) Total Marks: 1

Continuity of a function in a topological space depends on the collection of ..... of both sets.

Select the correct option

- open subsets
- improper subsets
- proper subsets
- closed subsets

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MTH634:Grand Quiz

Quiz Start Time: 04:00 PM, 02

Question # 12 of 30 (Start time: 04:12:04 PM, 02 July 2021)

Tota

_____ is the weakest topology on set X .

Select the correct option

Reload Math

- Indiscrete Topology
- Noe of above
- Cofinite Topology
- Discrete Topology

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Question # 25 of 30 (Start time: 04:09:40 PM, 02 July 2021)
Total Marks: 1

Function between two topological spaces is not continuous when the .....

Select the correct option

- Function is constant between any two topological spaces
- Domain space is indiscrete and range space is any space
- Domain space is discrete and range space is any space
- Domain space is any space and range space is indiscrete

Click to Save Answer & Move to Next Question

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MTH634:Grand Quiz

Quiz Start

Question # 29 of 30 (Start time: 04:45:35 PM, 02 July 2021)

Let X be a nonempty set and τ be a topology on X, then the following statements must be true. (*where* $P(X)$ *denotes* the power set of X)

Select the correct option

- $X \subset \tau$
- $\tau \subset X$
- none
- $\tau \subset P(X)$

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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 5 of 30 (Start time: 05:07:24 PM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d\}$.The following sets represents a topology on X .

Select the correct option

Reload Math Equations

- $\{\phi, \{a\}, \{b\}\}$
- $\{\{a\}, \{b\}, X\}$
- $\{\phi, \{a\}, \{a, b\}, \{c\}, X\}$
- $\{\phi, X\}$



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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 2 of 30 (Start time: 05:04:46 PM, 02 July 2021) **Total Marks: 1**

Consider the discrete topology D on $X=\{a,b,c\}$. Find a subbase B for D which does not contain any singleton sets.

Select the correct option

Reload Math Equations

- $B=\{\{a,b\}, \{b,c\}\}$
- $B=\{\{a,b\}, \{b,c\}, \{c,a\}\}$

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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 3 of 30 (Start time: 05:05:24 PM, 02 July 2021)

Total Marks: 1

..... of a subset A of a topological space (X, τ) is the set of all limit points of A.

Select the correct option

Reload Math Equations

○	All of above
○	Proper Set
○	Super Set
○	Derived Set

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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 4 of 30 (Start time: 05:06:24 PM, 02 July 2021)

Total Marks: 1

Consider the following class of subsets of $X=\{a, b, c, d\}$:

$$A=\{\{a, b\}, \{b, c\}, \{d\}\}$$

Which of the following is the topology on X generated by the class A.

Select the correct option

Reload Math Equations

- $T=\{ \emptyset, X, \{a, b\}, \{b,c\},\{b\}, \{d\}, \{a,b,d\}, \{b,c,d\}, \{b,d\}, \{a,b,c\}\}$
- $T=\{\emptyset, X, \{a, b\}, \{b,c\},\{b\}\}$

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Question # 7 of 30 (Start time: 05:09:05 PM, 02 July 2021) Total Marks: 1

If f(x)=x+2 , then f-1(a)=.....

Select the correct option

- a-2
- a+2
- 2/a
- 2a

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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 6 of 30 (Start time: 05:08:15 PM, 02 July 2021)

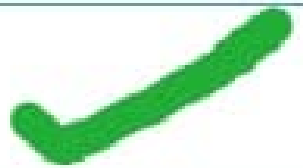
Total Marks: 1

A function $f : X \to Y$ from a topological space (X, T_X) to a topological space (Y, T_Y) is continuous iff for all $u \in T_Y$, there exist $v \in T_X$, such that

Select the correct option

Reload Math Equations

- $f^{-1}(v) = v$
- $f^{-1}(v) = u$
- $f^{-1}(u) = u$
- $f^{-1}(u) = v$



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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 8 of 30 (Start time: 05:09:54 PM, 02 July 2021)

Total Marks: 1

The smallest topology one can define on some set is called:

Select the correct option

Reload Math Equations

○	comparable topology
○	usual topology
○	discrete topology
○	indiscrete topology

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Question # 9 of 30 (Start time: 05:11:09 PM, 02 July 2021) Total Marks: 1

A function is continuous …… it is continuous at …… point of the domain space.

Select the correct option

- if and only if, every
- if and only if , any
- if, every
- if, any

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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 10 of 30 (Start time: 05:11:47 PM, 02 July 2021)

Total Marks: 1

The interior of the four intervals $[a,b]$, (a,b), $(a,b]$, $[a,b)$ is

Select the correct option

Reload Math Equations

- [a,b]
- (a,b)



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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 11 of 30 (Start time: 05:12:27 PM, 02 July 2021)

Total Marks: 1

Consider the following class of subsets of $X=\{a, b, c, d, e\}$:

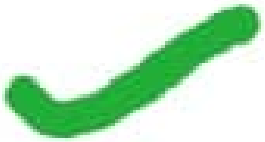
$$A=\{\{a, b, c\}, \{c, d\}, \{d, e\}\}$$

Which of the following is the topology on X generated by the class A.

Select the correct option

Reload Math Equations

- $T=\{\emptyset, X, \{c, d\}, \{d, e\}, \{c\}, \{d\}, \{c, d, e\}\}$
- $T=\{\emptyset, X, \{a, b, c\}, \{c, d\}, \{d, e\}, \{c\}, \{d\}, \{c, d, e\}, \{a, b, c, d\}, \{c, d, e\}\}$



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Question # 12 of 30 (Start time: 05:13:11 PM, 02 July 2021) Total Marks: 1

The basis for usual topology on R is the set of all .....

Select the correct option

- closed intervals
- open intervals 
- open disks
- closed disks

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MTH634:Grand Quiz

Quiz Start Time: 05:09 PM, 02 Ju

Question # 5 of 30 (Start time: 05:12:48 PM, 02 July 2021)

Total M

Which of the following is the smallest closed supe set of a subset A of a topological space X?

Select the correct option

- Complement of A
- Derived set of A
- Closure of a set A
- open sets of A

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Question # 13 of 30 (Start time: 05:13:54 PM, 02 July 2021)

Total Marks: 1

Which of the following topology contains the complete power set of a set?

Select the correct option

- lower limit topology
- Indiscrete topology
- co finite topology
- Discrete topology 

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Question # 14 of 30 (Start time: 05:15:24 PM, 02 July 2021)

Total Marks: 1

For any two topologies T1 and T2 defined on a non empty set then T1 is finer than T2 , iff for all U(open set) $\in$ T2 implies.....

Select the correct option

- $U \notin T1$
- $U \in T1$ 

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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 15 of 30 (Start time: 05:15:57 PM, 02 July 2021)

Total Marks: 1

An open disk in R^2 (Real plane) centered at $(x, y) \in R^2$, is the set of all points $(x', y') \in R^2$ such that

Select the correct option

Reload Math Equations

- $\{(x', y') | \sqrt{(x - x')^2 + (y - y')^2} = r$, where r is a positive real number$\}$
- $\{(x', y') | \sqrt{(x - x')^2 + (y - y')^2} > r$, where r is a positive real number$\}$
- $\{(x', y') | \sqrt{(x - x')^2 + (y - y')^2}\, a \leq r$, where r is a positive real number$\}$
- $\{(x', y') | \sqrt{(x - x')^2 + (y - y')^2} < r$, where r is a positive real number$\}$

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Question # 16 of 30 (Start time: 05:16:47 PM, 02 July 2021) Total Marks: 1

If X is a finite set then co-finite topology on X is .....

Select the correct option

- discrete topology
- none of these
- indiscrete topology
- lower limit topology

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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 18 of 30 (Start time: 05:19:21 PM, 02 July 2021)

Total Marks: 1

Let (X, τ) be the topological space then the following is true for closure of a set.

Select the correct option

Reload Math Equations

- $\overline{X} = X$
- $\overline{X} = \phi$
- $\overline{X} \subset X$
- None of above

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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 19 of 30 (Start time: 05:20:08 PM, 02 July 2021) **Total Marks: 1**

In a topological space, a subset of A of X closed if the compliment of A in X is open .

Select the correct option

Reload Math Equations

- True 
- False

Click to Save Answer & Move to Next Question

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Question # 15 of 30 (Start time: 05:21:12 PM, 02 July 2021)

Let $X = \mathbb{R}$ with usual topology and $B = [1, 2)$. Then interior of a set B is

Select the correct option

○	2
○	3/2
○	none
○	1

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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 21 of 30 (Start time: 05:21:53 PM, 02 July 2021)

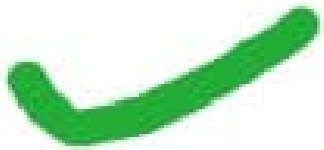
Total Marks: 1

In a topological space, the interior of a set A is the union of all

Select the correct option

Reload Math Equations

- closed subsets of A
- open subsets of A



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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 22 of 30 (Start time: 05:22:51 PM, 02 July 2021)

Total Marks: 1

Closure of A is the smallest closed superset of A.

Select the correct option

Reload Math Equations

- False
- True



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Question # 23 of 30 (Start time: 05:23:47 PM, 02 July 2021)

Total Marks: 1

(T1 Union T2) is the smallest topology containing .....

Select the correct option

- Only T1
- Only T2
- any one from T1 and T2
- both T1 and T2



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Question # 24 of 30 (Start time: 05:24:44 PM, 02 July 2021) Total Marks: 1

Every open subset of the real line is a union of disjoint .....

Select the correct option

- Semi Open Intervals
- Closed Intervals
- Open Intervals
- Sets

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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 25 of 30 (Start time: 05:25:24 PM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d\}$ and $\tau = \{\phi, \{a\}, \{a, b\}, X\}$ be a topology on X then the open sets of X are :

Select the correct option

Reload Math Equations

- $\{b\}, \{b, d\}$
- $\{a\}, \{a, b\}$
- $\{d\}, \{a, d\}$
- $\{c\}, \{c, d\}$

Click to Save Answer & Move to Next Question

Scanned with CamScanner

MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 26 of 30 (Start time: 05:26:56 PM, 02 July 2021)

Total Marks: 1

If τ is a topology on a non empty set X, then arbitrary ..... of members of τ belongs to τ.

Select the correct option

Reload Math Equations

- union
- intersection

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Question # 28 of 30 (Start time: 05:28:32 PM, 02 July 2021) Total Marks: 1

Which of the following statement is correct?

Select the correct option

- The composition of two continuous function may or may not continuous.
- The composition of two continuous function is a continuous function.
- The composition of two continuous function is piecewise continuous.
- The composition of two continuous function is discontinuous.

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MTH634:Grand Quiz

Quiz Start Time: 05:03 PM, 02 July 2021

Question # 29 of 30 (Start time: 05:29:11 PM, 02 July 2021)

Total Marks: 1

Let X ={a,b,c,d} with the topology $\tau = \{\phi, \{a\}, \{a,b\}, \{a,c,d\}, X\}$,then which of the following represents a subspace topology relative to A={a,b,c}?

Select the correct option

Reload Math Equations

- $\tau_A = \{\phi, \{a\}, \{a,b\}, A\}$
- $\tau_A = \{\phi, \{a\}, \{a,b\}, \{a,c\}, A\}$
- $\tau_A = \{\phi, \{a\}, \{a,c\}, A\}$
- $\tau_A = \{\phi, \{a\}, \{a,c,d\}, A\}$

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Question # 30 of 30 (Start time: 05:29:53 PM, 02 July 2021)

Total Marks: 1

A proper closed subset of a topological space X can never dense in X.

Select the correct option

- False
- True



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MTH634 Grand Quiz

Quiz Start Time: 05:09 PM, 02 July 202

Question # 25 of 30 (Start time: 05:30:42 PM, 02 July 2021)

Total Marks:

A function is continuous if and only if the inverse image of every ..... subset of Y under f is ..... in X.

Select the correct option

- Open, closed
- closed, open
- closed, closed
- Proper,closed

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